

3.2 Model-based decomposition

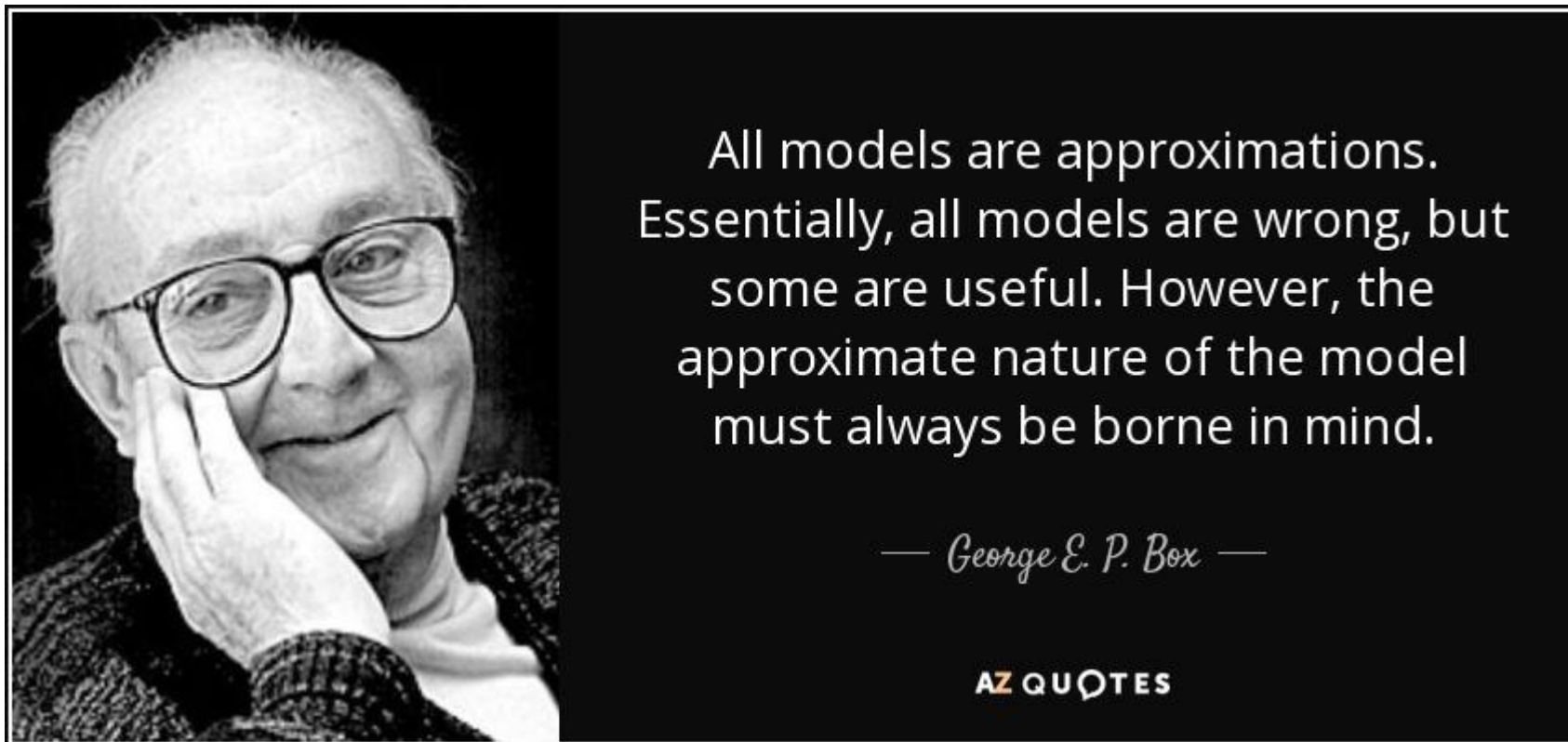
ESTP Training
05-07 October 2026

eurostat 

0. Outline

- Unobserved component models
 - STS, AMB
- SEATS algorithm
- Estimation
 - MMSE estimators, algorithms
- Analysis
 - Finite / infinite approaches
- SEATS in JD+
 - Parameters, non decomposable models, estimation

Always keep in mind...



0.1 Some notations

- Operators

- Backward, forward operator: $B^a y_t = y_{t-a}, F^a y_t = y_{t+a}$
- Differencing: $\nabla_f y_t = y_t - y_{t-f} = (1 - B^f) y_t$
- Annual sum: $S_f y_t = y_t + \dots + y_{t-f+1}$
- $\nabla_f = \nabla_1 S_f$

- ARMA

- $\Phi(B) y_t = \Theta(B) \varepsilon_t$

- ARIMA

- $\Delta(B) \Phi(B) y_t = \Theta(B) \varepsilon_t$

0.2 Useful tools

- Wold representation of an ARMA model

$$\Phi(B)y_t = \Theta(B)\varepsilon_t \Leftrightarrow y_t = \Psi(B)\varepsilon_t, \quad \Psi(B) = \frac{\Theta(B)}{\Phi(B)}$$

- Auto-covariance generating function (acgf)

$$acgf(y_t) = \sigma^2 \Psi(z^{-1})\Psi(z) = \sigma^2 \frac{\Theta(z^{-1})\Theta(z)}{\Phi(z^{-1})\Phi(z)}$$

- Example

- $y_t = (1 + .8B)\varepsilon_t \Leftrightarrow \Psi(B) = 1 + .8B$

- $acgf(y_t) = (1 + .8z^{-1})(1 + .8z) = 1.64 + .8z^{-1} + .8z$

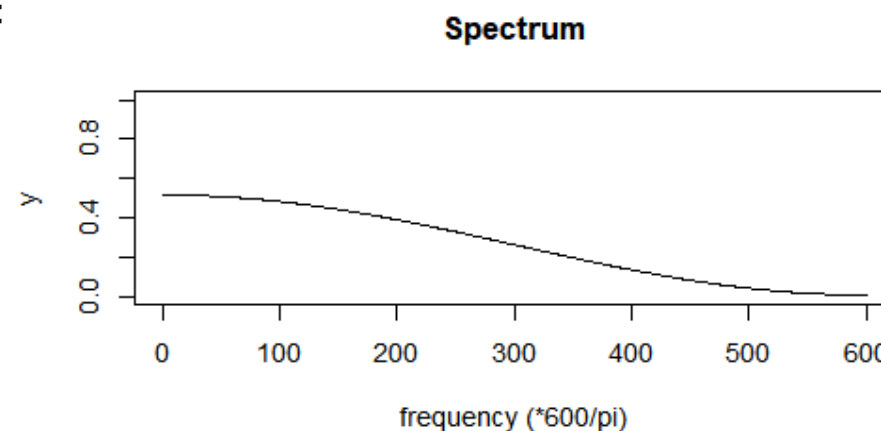
0.2 Useful tools

- Spectrum $\equiv$ Fourier transform of the acgf

$$g_y(\lambda) = \frac{\sigma^2 |\Theta(e^{i\lambda})|^2}{2\pi |\Phi(e^{i\lambda})|^2}$$

- Example (cont.)

$$\begin{aligned} g_y(\lambda) &= \frac{1}{2\pi} |1 + .8 \cos \lambda + .8 i \sin \lambda|^2 = \\ &= \frac{1}{2\pi} ((1 + .8 \cos \lambda)^2 + (.8 \sin \lambda)^2) \\ &= \frac{1}{2\pi} (1.64 + 1.6 \cos \lambda) \end{aligned}$$

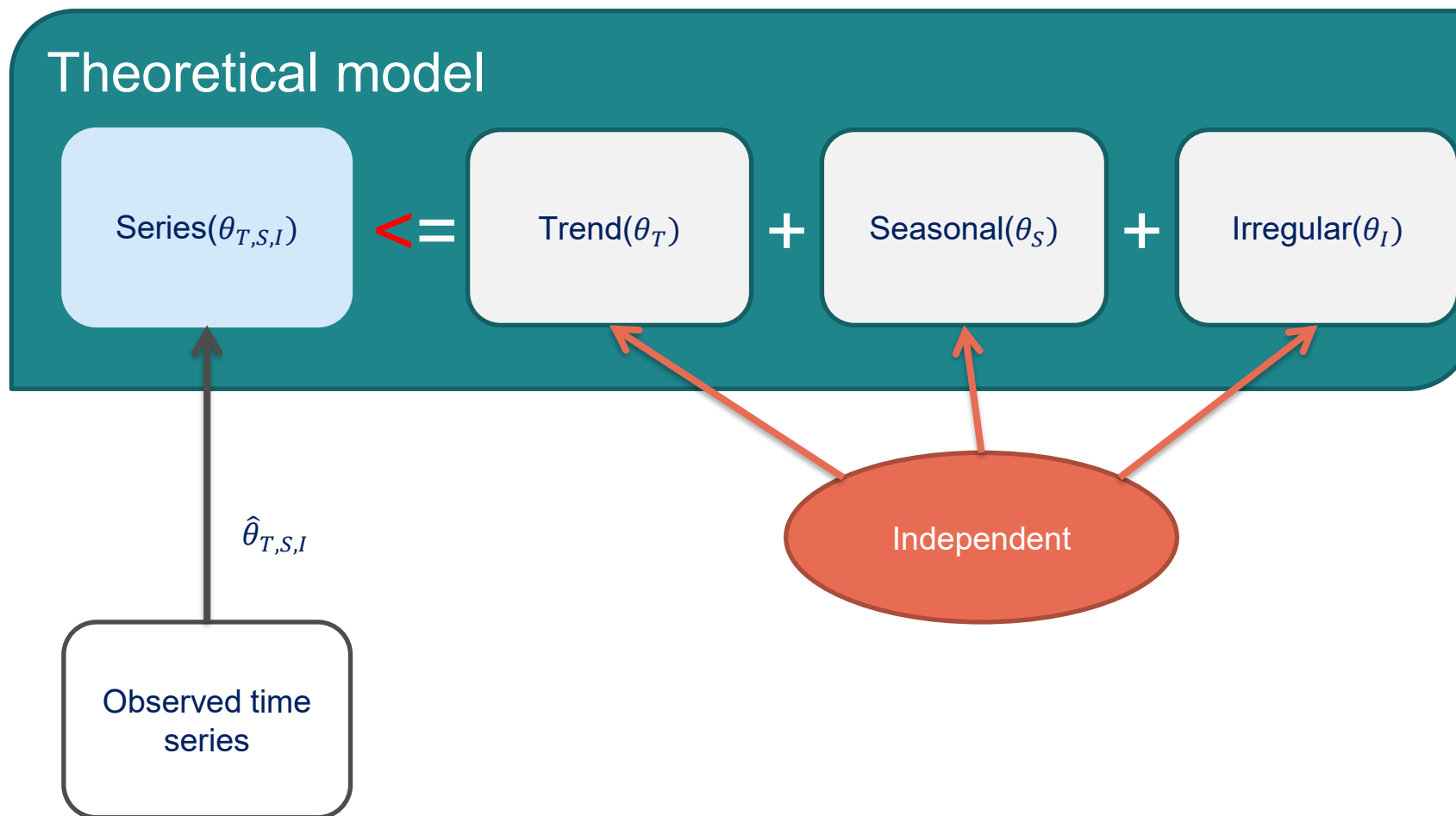


- Formal extension to the non-stationary cases

1. Unobserved components model

- Principle
 - The different components of the series are “defined” by [ARIMA] models
 - The components are independent
 - The sum of the model of the components is the model of the series

1.1 Unobserved components model (STS)



1.1.1 Basic structural model (STS)

- Local linear trend (l_t)

$$\Delta l_t = m_t + \mu_t, \quad \mu_t \sim N(0, \sigma_\mu^2)$$

$$\Delta m_t = v_t, \quad v_t \sim N(0, \sigma_v^2)$$

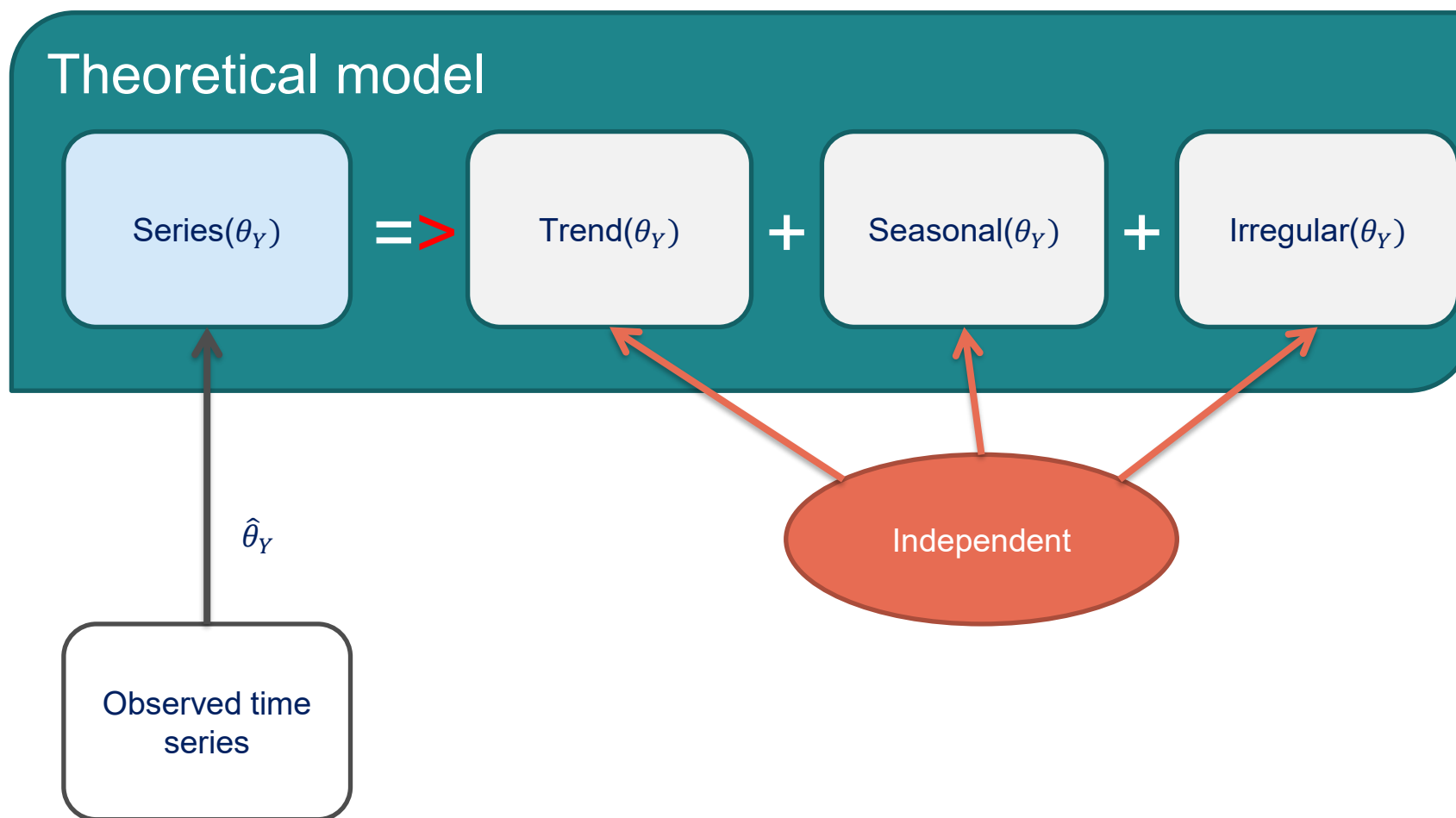
- Seasonal component (i_t)(*dummy model*)

$$S_f s_t = \gamma_t, \quad \gamma_t \sim N(0, \sigma_\gamma^2)$$

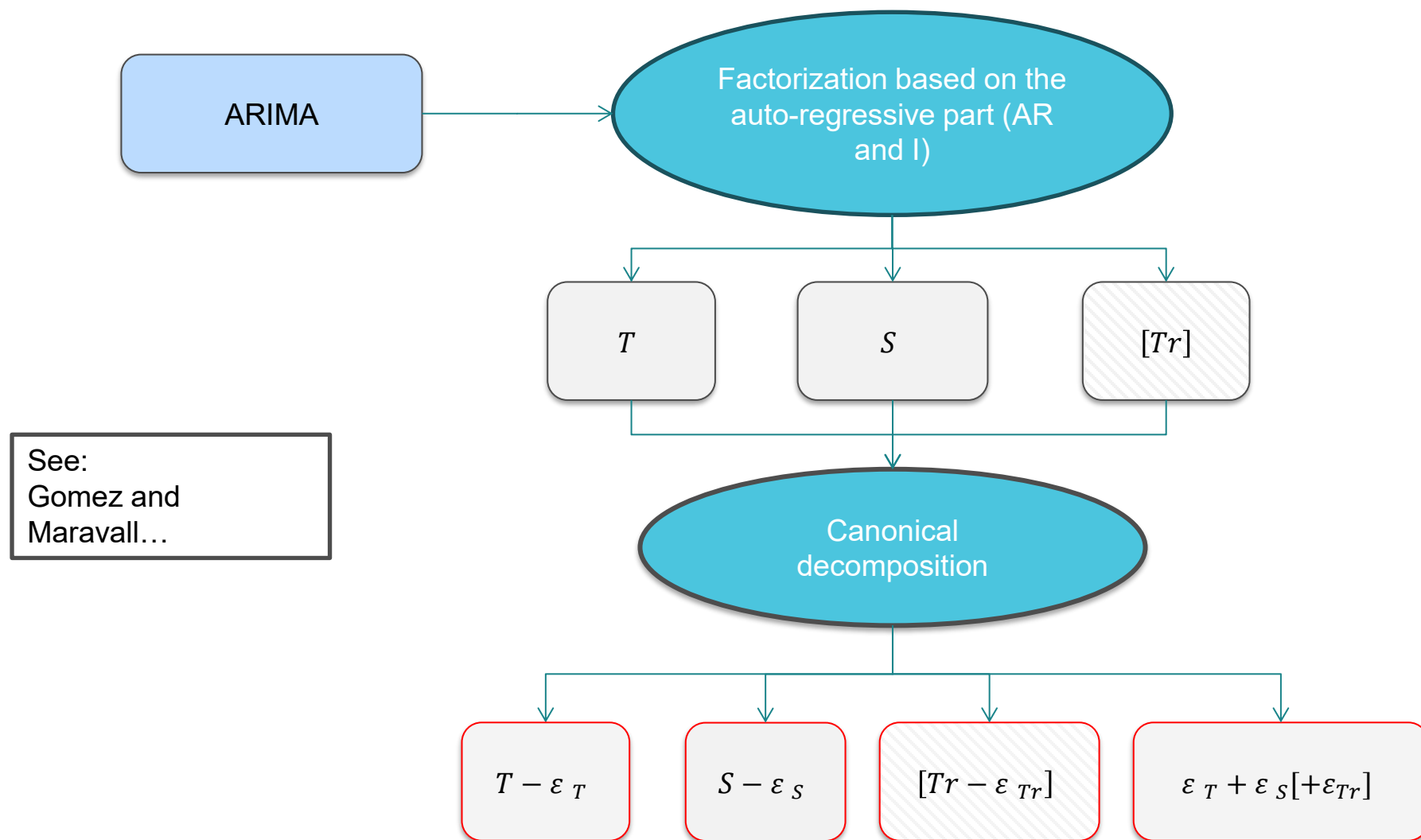
- Noise (i_t)

$$i_t = \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

1.2 Unobserved components model (AMB)



2.1 SEATS algorithm



2.2 SEATS Algorithm. Details (I)

- ARIMA:

$$\Phi(B)\Delta(B)y_t = \Theta(B)\varepsilon_t$$

- Factorization of the AR polynomial:

$$\Phi(B)\Delta(B) = \prod (1 - \alpha_i B)$$

- Trend-cycle

- α_i real,
 - $\alpha_i \geq k$
- α_i complex,
 - $|\alpha_i| \geq k, \arg(\alpha_i) \leq c$

$$\left(c = \frac{\pi}{s}\right) \sim \text{cycle length} \geq \text{two years}$$

TrendCycleSelector.java

```
public boolean accept(final Complex root) {
    Complex iroot = root.inv();
    if (root.getIm() == 0) {
        return iroot.getRe() >= m_bound;
    } else {
        if (iroot.abs() >= m_bound) {
            double arg = Math.abs(iroot.arg());
            if (arg <= m_lfreq) {
                return true;
            }
        }
        return false;
    }
}
```

- Seasonal

- α_i real,
 - $\alpha_i < -l$
 - α_i complex,
 - $|\arg(\alpha_i) - f_s| \leq e$

f_s seasonal frequency

SeasonalSelector.java

```
public boolean accept(final Complex root) {
    if (Math.abs(root.getIm()) < 1e-6) {
        if (1/root.getRe() < -m_k)
            return true;
        else
            return false;
    }

    double pi = 2 * Math.PI / m_freq;
    double arg = Math.abs(root.arg());
    double eps = m_epsphi / 180 * Math.PI;
    for (int i = 1; i <= m_freq / 2; ++i) {
        if (Math.abs(pi * i - arg) <= eps)
            return true;
    }
    return false;
}
```

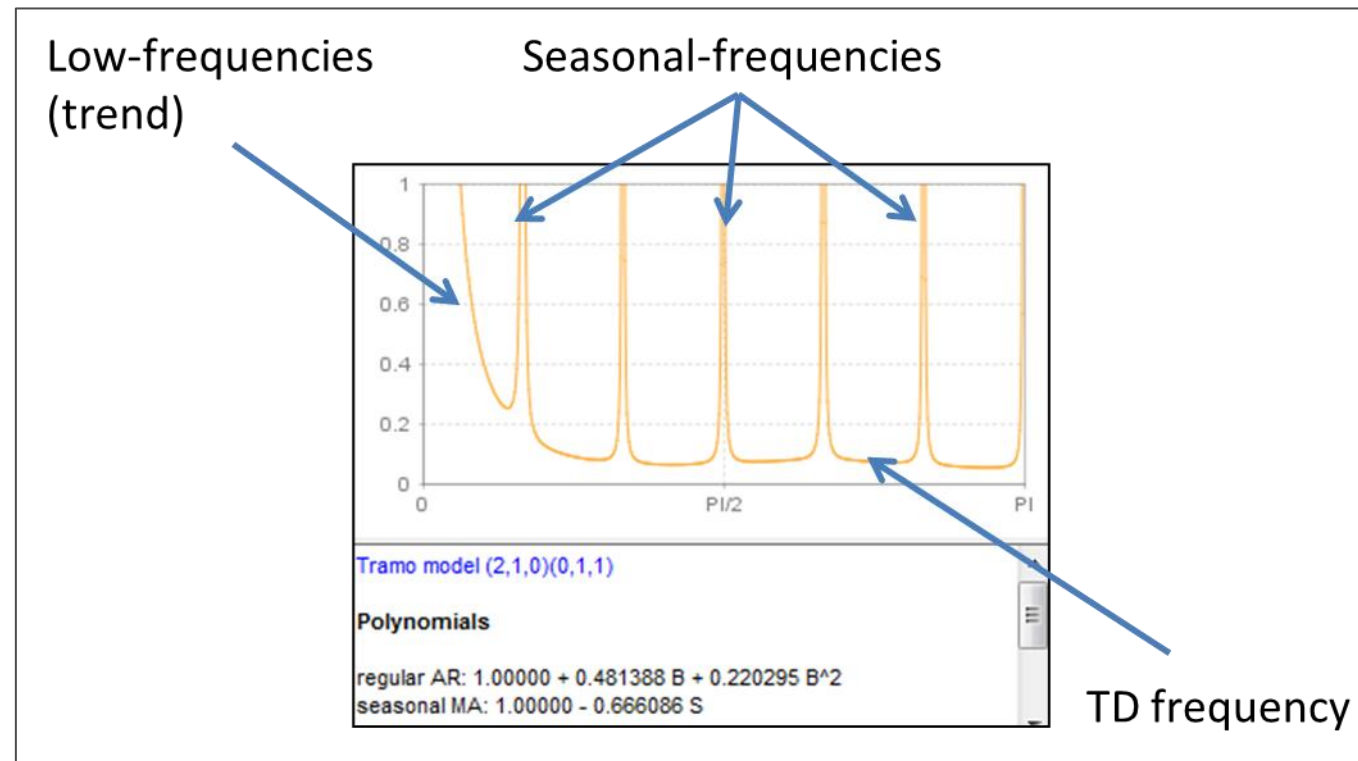
- Transitory (I)

- All other roots

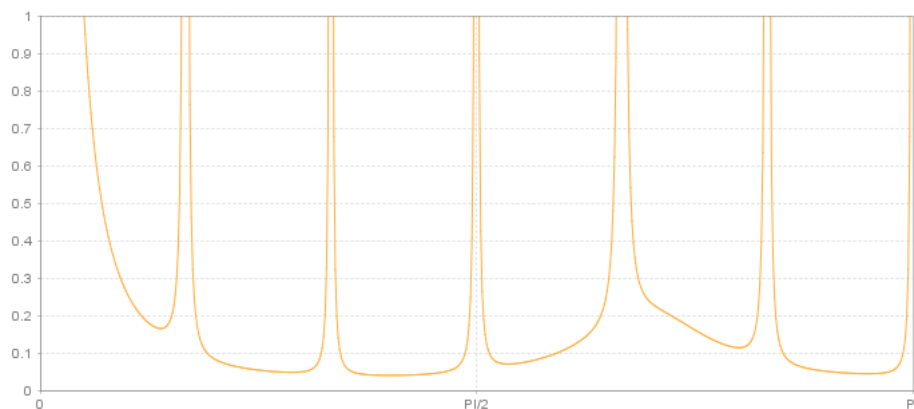
Trend boundary	k
Seasonal tolerance (degree)	e
Seasonal boundary	l
Seas.boundary (unique)	l (no seasonal part)

2.3 SEATS = spectral decomposition

- Splitting of the spectrum following the different « types » of frequencies



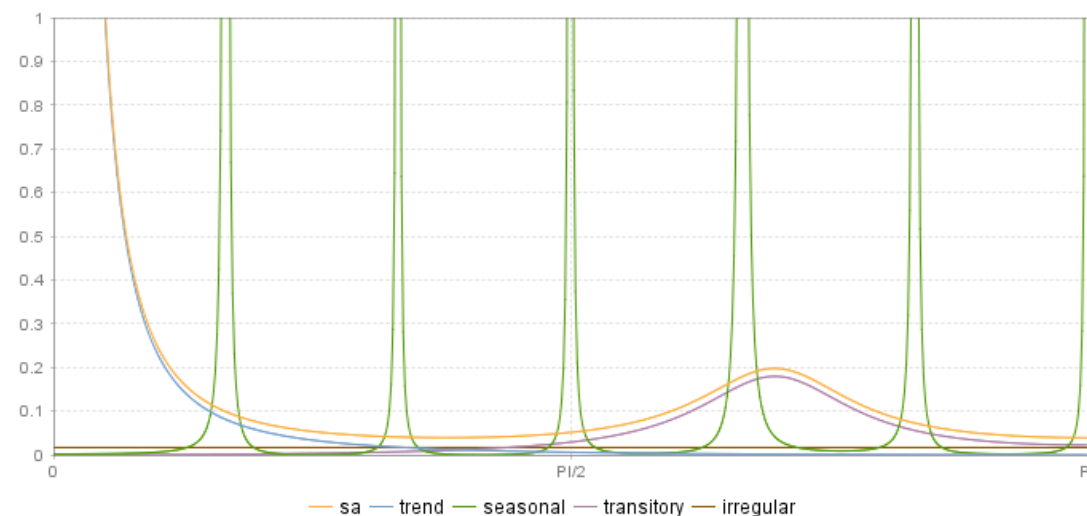
2.3 SEATS = spectral decomposition



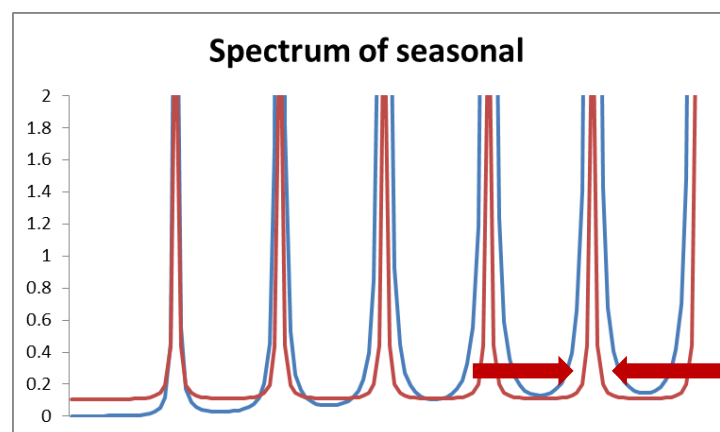
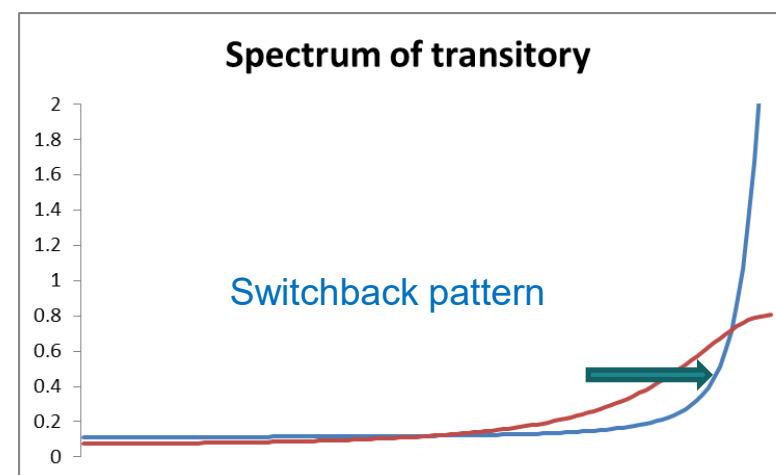
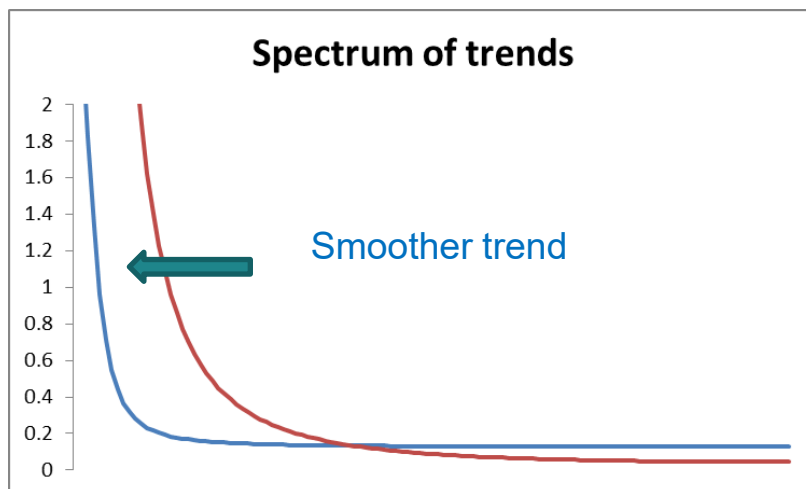
$$S_Y = S_T + S_S + S_{Tr} + S_I$$

$$S_{SA} = S_T + S_{Tr} + S_I = S_Y - S_S$$

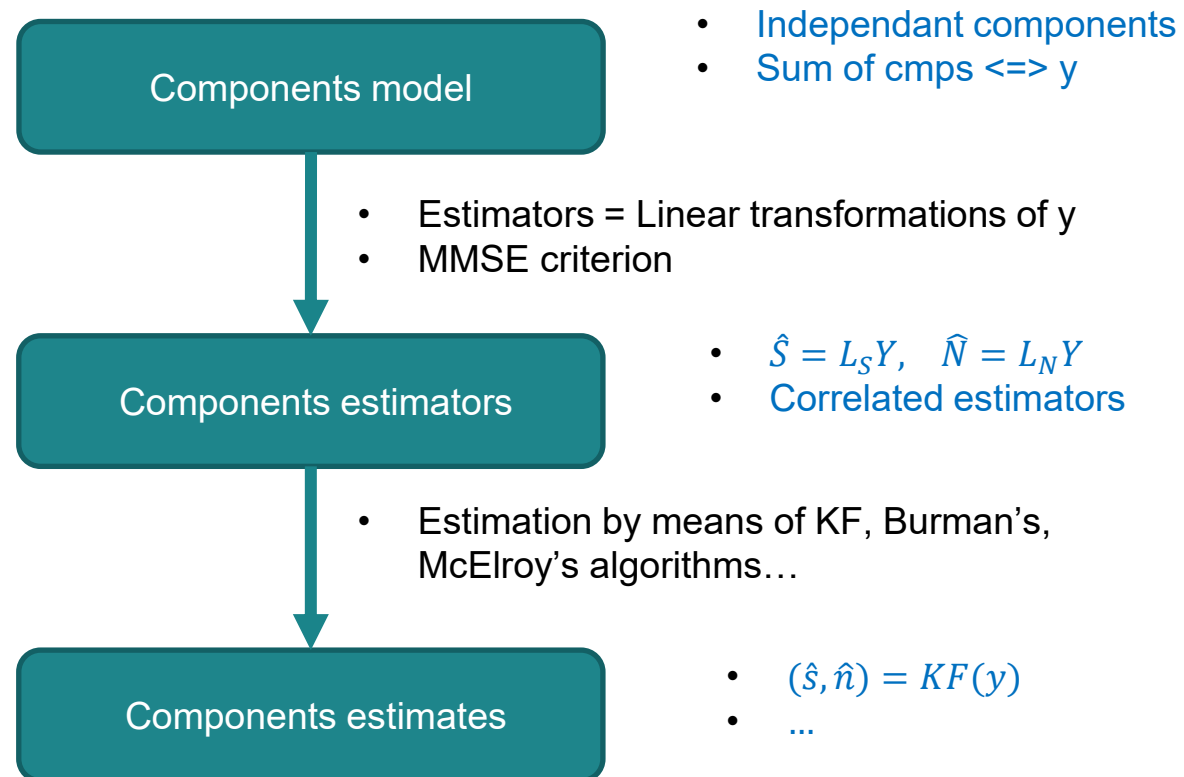
Independent components !



2.3 SEATS = spectral decomposition



3. Estimation of the model



3.1 Estimation of the components

- STS and AMB → UCARIMA models
 - Same estimation algorithms
 - Common analysis tools
- 3 solutions, strictly equivalent (except for SD)
 - Burman algorithm (WK filters): legacy and default solution, fastest, rich analysis framework, (semi-)infinite approach
 - Kalman smoother: more stable but complex, exact finite properties
 - [Matrix computation]: Not available in JD+ 3.x



3.1.1 Components

- $\Phi(B)y_t = \Theta(B)\varepsilon_t$
- $\Phi_s(B)s_t = \Theta_s(B)\varepsilon_{s,t}$
- $\Phi_n(B)n_t = \Theta_n(B)\varepsilon_{n,t}$

- $y_t = n_t + s_t$
 $\Downarrow$
- $\Phi_y(B) = \Phi_s(B)\Phi_n(B)$
- $\Theta_y(B)\varepsilon_t = \Phi_n(B)\Theta_s(B)\varepsilon_{s,t} + \Phi_s(B)\Theta_n(B)\varepsilon_{n,t}$

3.1.2 Estimators

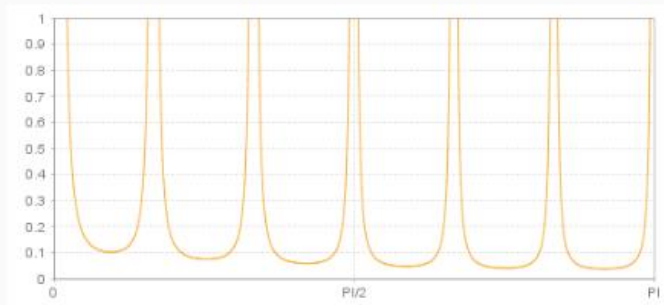
MMSE criterion



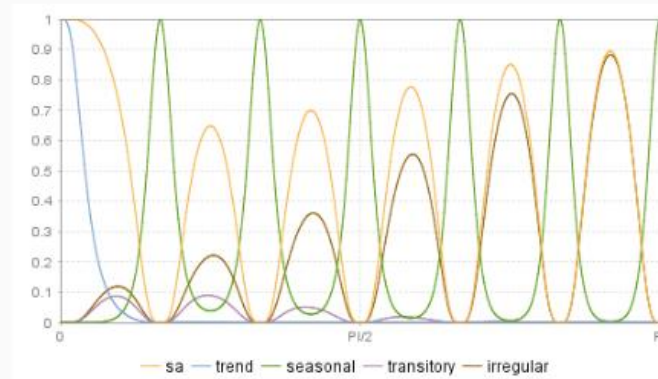
Wiener-Kolmogorov filter (Whittle, 1963)

- $\hat{s}_t = k_s \left[\frac{\Theta_s(B)\Phi_n(B)\Theta_s(F)\Phi_n(F)}{\Theta(B)\Theta(F)} \right] y_t$
- $\hat{s}_t = k_s \left[\frac{\Theta_s(B)\Phi_n(B)\Theta_s(F)\Phi_n(F)}{\Theta(B)\Theta(F)} \right] \left[\frac{\Theta(B)}{\Phi(B)} \right] \varepsilon_t = k_s \left[\frac{\Theta_s(B)\Theta_s(F)\Phi_n(F)}{\Phi_s(B)\Theta(F)} \right] \varepsilon_t$
- $\hat{s}_t = \sum_{i \leq t} \xi_{s,i}^- \varepsilon_i + \sum_{i > t} \xi_{s,i}^+ \varepsilon_i$
- $\hat{s}_{t|t} = \sum_{i \leq t} \xi_{s,i}^- \varepsilon_i$ (concurrent estimator) , $r_t = \sum_{i > t} \xi_{s,i}^+ \varepsilon_i$ (revisions)

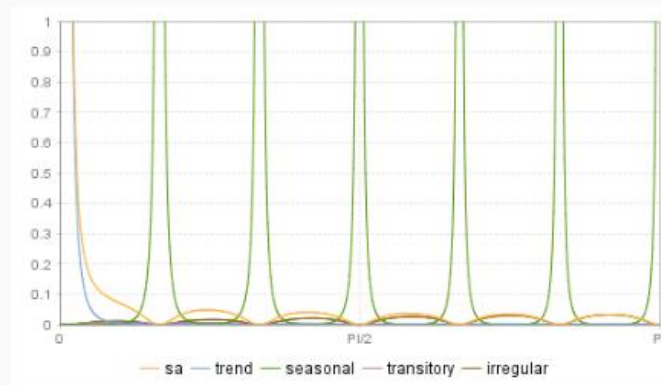
4.1 Analysis : Graphical representations



(a) $g(\omega)$



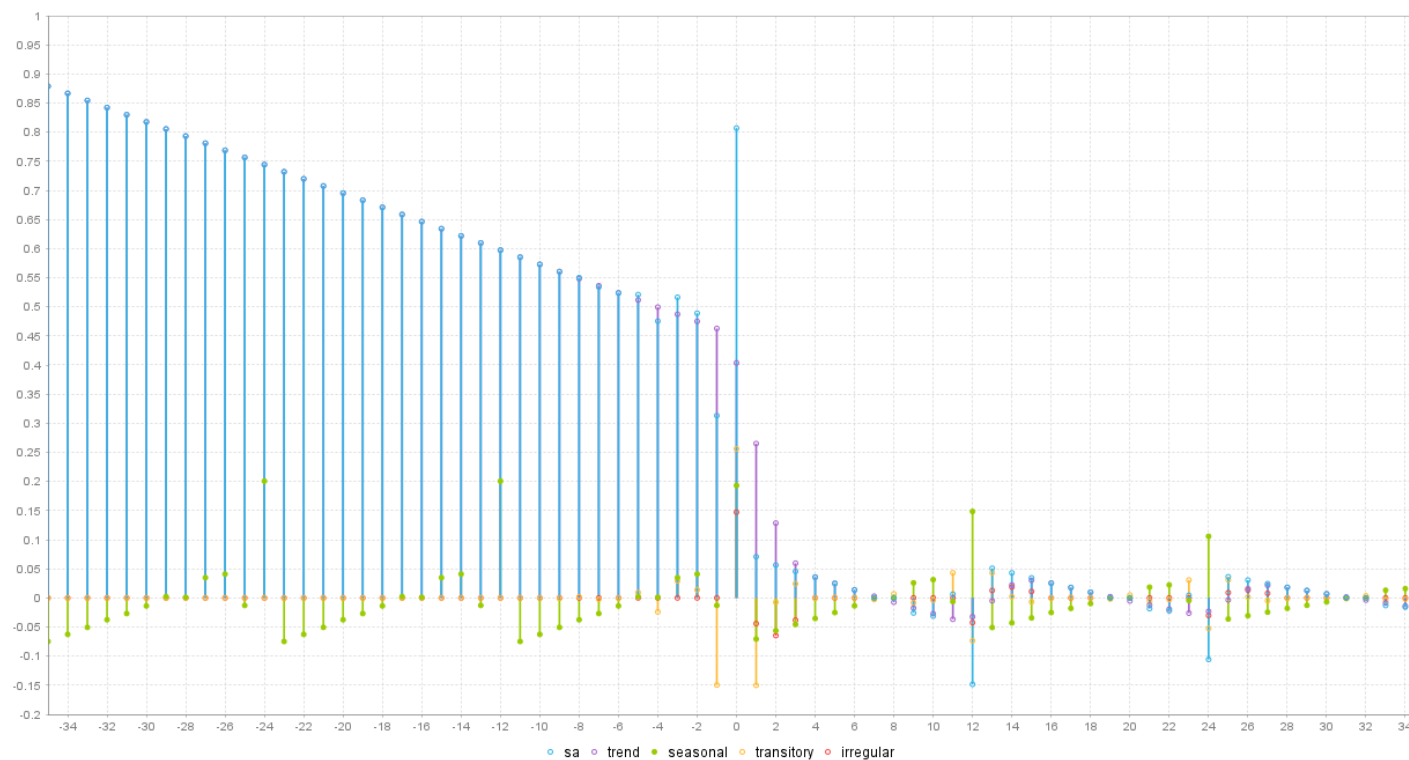
(b) $\nu_X^2(\omega)$



(c) $g_{\hat{X}}(\omega)$

4.1 PsiE-weights

- =Psi weights for estimators



4.1 Properties of the components, estimators, estimates

Distribution of component, theoretical estimator and empirical estimate (stationary transformation)

Variance

	Component	Estimator	Estimate	P-Value
Trend	0.0060	0.0001	0.0001	0.3219
Seasonally adjusted	1.4296	0.8100	0.7529	0.6307
Seasonal	0.6917	0.3057	0.2453	0.3731
Irregular	0.2967	0.1449	0.1308	0.3682

Autocorrelation

Trend

Lag	Component	Estimator	Estimate	P-Value
1	0.0036	0.6011	0.6216	0.6546
2	-0.4964	0.0708	0.1252	0.5681
3	0.0000	-0.0153	0.0821	0.3281
4	0.0000	0.0296	0.1949	0.0986
5	0.0000	0.0288	0.1484	0.2000
6	0.0000	0.0278	-0.0362	0.3585
7	0.0000	0.0267	-0.1486	0.0589
8	0.0000	0.0255	-0.1141	0.1582
9	0.0000	0.0415	-0.0615	0.2927
10	0.0000	0.0076	-0.1131	0.2189
11	0.0000	-0.1956	-0.2827	0.3205
12	0.0000	-0.3488	-0.3778	0.6880

Seasonally adjusted

Lag	Component	Estimator	Estimate	P-Value
1	-0.6263	-0.6421	-0.6869	0.3063
2	0.1103	0.1339	0.2599	0.1876
3	0.0126	0.0059	-0.1429	0.1521
4	0.0027	0.0018	0.0672	0.5279
5	0.0006	0.0004	0.0732	0.4357
6	0.0001	0.0001	-0.1026	0.1504
7	0.0000	-0.0001	0.0578	0.5349
8	0.0000	-0.0007	-0.0867	0.4062
9	0.0000	-0.0022	0.1425	0.1630
10	0.0000	-0.0510	-0.1872	0.1818
11	0.0000	0.2446	0.3076	0.4713
12	0.0000	-0.3809	-0.4096	0.6872

4.2 Analysis

- What matters?
 - Understanding the differences between the “theoretical components” and their “estimators”
 - For instance: “dips” in the spectrum of the estimator
 - Understanding the properties of the estimators
 - Model of Irregular $\neq$ white noise, negative $ac(1)$ in many SA estimators, ...
 - Understanding PsiE-weights
 - Revisions of the components = forecasts errors * PsiE weights
 - Differences between the “final” estimator and the “concurrent” estimator
- Diagnostics: comparison of the properties of the estimators and of their estimates (variance, auto-correlations...)

5. SEATS in JD+

The screenshot shows the SEATS configuration window with the following parameters and values:

SEATS	
Approximation mode	Legacy
MA unit root boundary	0.95
Trend boundary	0.5
Seasonal tolerance	2
Seasonal boundary	0.8
Seas. boundary (unique)	0.8
Method	Burman
BENCHMARKING	
Is enabled	KalmanSmoother
Target	McElroyMatrix

Annotations:

- Modification of the model (points to the top section)
- Decomposition of the model (points to the middle section)
- Estimation of the components (points to the bottom section)

5.1 SEATS in JD+ (cont.)

- Impact of the parameters
 - **Only in case of AR polynomials**
 - k
 - Small: possible « noisy » trend
 - $k \approx 1$: more stable trend
 - e
 - Large (>5): possible short term cycle in the seasonal (for instance, stochastic TD)→erratic seasonal
 - l
 - Small ($<.8$): higher risk of erratic seasonal
- General consideration: threshold effects are unavoidable



5.2 Non decomposable models

- Some models are not decomposable (often due to complex models)
 - Best solution: change yourself the model ($\rightarrow$ airline)
 - Otherwise:
 - Legacy: Seats search for another SARIMA model, as similar as possible to the original model
 - Noisy: Seats add noise in the initial model ($\rightarrow I = Tr[= 0]$)

6. Final remarks

- What matters?
 - Impact of the model on
 - SA/S “smoothness” $\Rightarrow$ Be very careful with stationary AR roots
 - Revisions
 - Use PsiE-weights to understand/anticipate revisions
 - Model-based diagnostics
 - Variance estimators $\leftrightarrow$ variance estimates
 - Should not happen if the original model is well defined
 - Be careful with non decomposable models / fixed models / “bad” models
 - To go further, see:
 - “SEASONAL ADJUSTMENT AND SIGNAL EXTRACTION IN ECONOMIC TIME SERIES”, by R. Gomez and A. Maravall

6. Final remarks (cont.)

- Advantages and drawbacks of model-based decomposition
 - +
 - « Optimal filters », tailored for each series
 - Many useful additional information (uncertainty...)
 - -
 - Might generate huge revisions
- Some recommendations
 - Fix the model
 - Fix the series in the past (5 years revisions)
 - Fix the parameters and revise them when you revise the raw data
 - Be pragmatic !

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